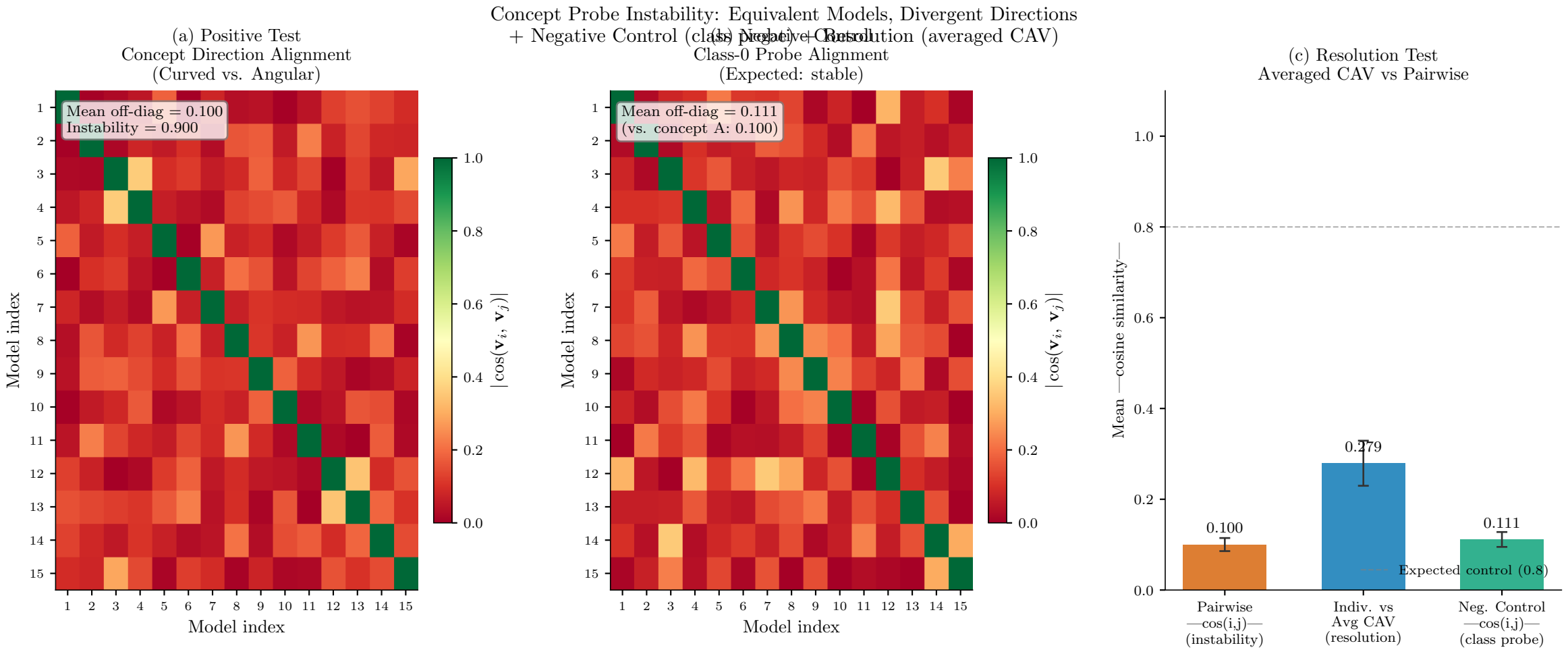
Concept Probe Instability: Equivalent Models, Divergent Directions
+ Negative Control (class probe) + Resolution (averaged CAV)
(a) Positive Test
Concept Direction Alignment
(Curved vs. Angular)
Mean off-diag = 0.100
Instability = 0.900
Model index
Model index
$|\cos(\mathbf{v}_i, \mathbf{v}_j)|$
(b) Negative Control
Class-0 Probe Alignment
(Expected: stable)
Mean off-diag = 0.111
(vs. concept A: 0.100)
Model index
Model index
$|\cos(\mathbf{v}_i, \mathbf{v}_j)|$
(c) Resolution Test
Averaged CAV vs Pairwise
Mean |cosine similarity|
0.100
0.279
0.111
Expected control (0.8)
Pairwise |cos(i,j)| (instability)
Indiv. vs Avg CAV (resolution)
Neg. Control |cos(i,j)| (class probe)